Assignment 1: Auditing and Explaining Diabetes Risk Prediction Models

Group size: 4 students
Presentation: Week 7 (20.10.2026)
Report hand-in: 24.10.2026

Brief description: This assignment bridges the gap between technical explainability (XAI) methods and user experience (UX). You will train and compare different machine learning models, use XAI to "audit" your own creations for systematic errors, and practice the art of translating complex model behaviors into tailored visual explanations for different stakeholders.

The Scenario

You have been hired as an external data science consulting team by DiabetesPatientsRuleForever, an interest organization for Diabetes patients. They want to give people at risk of developing diabetes increased insight into their disease risk. They therefore want you to create an AI model to automatically predict diabetes risk, and use XAI to give people personalized feedback about what causes increased and decreased risk of developing diabetes.

They need you to develop the best possible model for this task, but they also need XAI feedback to be correct and safe – incorrect feedback to people at risk of developing disease could lead to them making lifestyle choices that put them at increased risk.

DiabetesPatientsRuleForever has hired you to:

1. **Develop** at least two different candidate risk prediction models using different algorithms.
2. **Use** XAI methods to provide feedback on why the model gives a particular risk, as well as what might cause a higher or lower future risk
3. **Validate** the performance of your explanations, and critically discuss risks and opportunities associated with using them for this purpose
4. **Explain** your chosen model to key stakeholders using appropriate visualizations.

Dataset: We will use the version of the CDC BRFSS Diabetes dataset named diabetes_binary_5050split_health_indicators_BRFSS2015.csv (please find the file on Learn). This dataset is 50-50 split between healthy subjects and patients with prediabetes or diabetes, and its features are explained by the BRFSS 2015 codebook. This dataset comes with the task of predicting whether or not subjects suffer from (pre)diabetes.

Models: You must train **at least two distinct machine learning models** (e.g., logistic regression, random forest, decision tree, or neural network).

The audit: You must use **at least two different XAI methods** (e.g., permutation feature importance, LIME, SHAP, gradient-based methods, LRP) to analyze your models. You must design static visual explanations of **both models for three distinct stakeholders** (see below). You cannot use the

same visualization for all of them; the content must be tailored to their specific needs. Do the explanations help you discover model failure modes? Can the explanations be helpful for patients?

Bonus (optional): If you discover significant errors, you are encouraged (but not required) to try to fix them. Even if you cannot fix them, you must clearly acknowledge and explain the risks in your report.

The stakeholders:

1. Head of data science at consultancy company
   Goal: debugging and validation.
   Needs: High fidelity, global feature importance, error analysis, edge cases.
2. Director of patient organization (non-technical)
   Goal: risk assessment and business logic.
   Needs: "Can we trust this?" High-level logic, proof of reliability, no math jargon.
3. Patient (end user)
   Goal: understand how to improve their risk of developing diabetes.
   Needs: "Why was I predicted to have an increased risk of diabetes?" and "What can I change to avoid getting diabetes?" (actionability).

Presentation (role-play)

Format: 10-minute presentation + 10-minute Q&A/feedback.
Rules:

- You must prepare explanation materials for all three stakeholders.
- On the day of the presentation, we will randomly assign your group one of the three stakeholders. You will not know which one until you stand up to present.
- The Audience: Another student group will act as that specific stakeholder. They will be given a feedback form and must ask at least 3 critical questions based on their role (e.g., if they are the patient: "I don't understand this chart. What do I need to do to stay healthy?").

Written report

Format: max 5 pages of text + illustrations/figures in appendix.
Submission: DTU Learn
Required structure:

1. Methodology:
   - Brief details of the two (or more) machine learning algorithms you chose to train and why. How do they differ?
   - If you attempted to improve the models, describe your approach and its results.
2. Justification of the XAI methods:
   - Why did you choose these specific methods (e.g., LIME vs. SHAP)?
   - Argument for why they are appropriate for this specific data type and these stakeholders.
3. UX and stakeholder analysis:
   - Explicitly define the users: Who are they? What are their goals? What decisions do they make based on the AI? What could go wrong?

- Describe your "Conceptual Model".

4. Visualizations and analysis:
    - Present the static visualizations you created.
    - Do the models rely on ethical features? Do they contain bias?
    - Analyze why they differ (e.g., How did you reduce cognitive load for the director? How did you add detail for the data scientist?)
5. Role-play reflection:
    - Briefly document the feedback you received during the presentation.
    - Did the "User" understand your explanation? What would you improve?
    - Based on those roleplays that you watch: Name 3 insights you got from the other presentations that you plan to use/change in your own project.